Stack Implementation Using Array

Program:

```
#include<stdio.h>
int stack[100], n, top, i;

void push(int x)
{
if(top >= n-1)
{
printf("\n STACK is over flow");
 {
else
{
top++;
stack[top] = x;
printf("\n element pushed into stack:%d",x);
}

{ void pop()
if(top <= -1) {
printf("\n Stack is under flow");

else
{
printf("\n The popped elements is %d", stack[top]);
stack[top] = 0;
top--;

void display()
{
if(top >= 0) {
printf("\n The elements in STACK:\n");

for(i=top; i>=0; i--)
printf("%d \n",stack[i]);
}
else
{
printf("\n The STACK is empty");
 }
}
```

```
int main()
{
top = -1; n = 5;
push(10);
push(20);
push(30);
push(40);
push(50);
push(60);
display();
pop();
pop();
pop();
pop();
pop();
pop();
pop();

return 0;
}
```

Output:

```
element pushed into stack:10
element pushed into stack:20
element pushed into stack:30
element pushed into stack:40
element pushed into stack:50
STACK is over flow

The elements in STACK:
50
40
30
20
10

The popped elements is 50
The popped elements is 40
The popped elements is 30
The popped elements is 20
The popped elements is 10
Stack is under flow
```